

Exploration of 100 commonly used drugs and supplements on cognition in older adults

Background There are conflicting reports and a lack of evidence-based data regarding effects of medications on cognition in cognitively normal older adults. We explored whether use of 100 common medications taken by older adults is associated with longitudinal cognitive performance. **Methods** A longitudinal observational cohort was used with analysis of data collected September 2005 through May 2011 and maintained in the National Alzheimer's Coordinating Center (NACC) Uniform Data Set. Participants were aged 50 years or older and cognitively normal (N=4414). Composite scores were constructed from 10 psychometric tests. Scores for each participant reflecting change in the psychometric composite score from the baseline clinical assessment to the next assessment were calculated. General linear models were used to test whether the mean composite change score differed for participants who reported starting, stopping, continuing, or not taking each of the 100 most frequently-used medications in the NACC sample. **Results** The average time between assessments was 1.2 years (SD=0.42). Nine medications showed a difference ($p<0.05$) across the four participant groups in mean psychometric change scores from the first to the second assessment. Medications associated with improved psychometric performance were: naproxen, calcium-vitamin D, ferrous sulfate, potassium chloride, flax, and sertraline. Medications associated with declining psychometric performance were: bupropion, oxybutynin, and furosemide. **Conclusions** Reported use of common medications is associated with cognitive performance in older adults, but studies are needed to investigate the mechanisms underlying these effects.